

CSE 260

Assignment 1

Marks : $5 \times 5 = 25$

Deadline : 6th July 2025 , 11:59 PM (Sunday)

N.B. Show every calculation step.

1	You are working as a cashier in a tech store. A customer buys a monitor priced at \$750 and returns an item worth \$215. Use 2's complement in 12 bits to calculate how much the customer owes now.	5
2	A delivery robot stores its travel coordinates using Excess-3 binary format. The robot reports its current position as: (010011100001.01101100)Excess-3. Convert this to the original coordinate values in decimal so the control center can decode it. (If the answer shows error, explain that error only)	5
3	In a game development system that uses base-6 for scoring, your character earns $(45)_6$ points in one level and $(23)_6$ in another. Multiply these two values using base-6 multiplication, and provide the final score in base-6.	5
4	You are designing a network simulation where time is measured in base-5 units. A task takes $(243)_5$ time units, and each subtask takes $(14)_5$ units. How many subtasks can be done, and what will be the remainder? Use base-5 division and show the complete working process.	5
5	You're building a custom PC: A 1TB SSD costs $(7F)_{16}$ dollars. A processor costs $(11010100)_2$ dollars. Your friend gives you $(2374)_8$ dollars.	5

	How much money will you have left after buying both components?	
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